

■ A.2.2 Continued Fractions

In number theory and dynamical systems theory it is often convenient to represent numbers as “continued fractions”, of the form $\frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots}}}$. The following *Mathematica* package takes a number, and generates a list of the quantities a_i in the continued fraction expansion. To show what kind of object the resulting list is, the package tags it with the head `ContinuedFractionForm`.

```
ContinuedFraction[x_Real, n_Integer?Positive] :=
  Block[ {
    xi, xp = x, r = {}
  },
  Do[
    xi = Floor[xp] ;
    AppendTo[r, xi] ;
    xp = 1 / (xp - xi) ,
    {n} ] ;
  Return [ ContinuedFractionForm[r] ]

ContinuedFractionForm/: Normal[ContinuedFractionForm[a_List]] :=
  Block[ {
    x, ar = Reverse[a]
  },

  x = ar[[1]] ;
  Do [ x = 1/x + ar[[i]], {i, 2, Length[ar]} ] ;
  Return [ x ]
```

Continued fractions.

Read in the continued fractions package.

```
In[1]:= <<ContinuedFractions.m
```

Here is $\sqrt{7}$ to 40 decimal places.

```
In[2]:= N[Sqrt[7], 40]
Out[2]= 2.64575131106459059050161575363926042571
```

This finds the first twenty continued fraction coefficients. Notice that the sequence becomes periodic, as it does for any square root.

```
In[3]:= ContinuedFraction[%, 20]
Out[3]= ContinuedFractionForm[{2, 1, 1, 1, 4, 1, 1, 1, 4,
1, 1, 1, 4, 1, 1, 1, 4, 1, 1, 1}]
```

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The package also specifies that when `Normal` is applied to a `ContinuedFractionForm`, it is converted to an ordinary number. This is a rational number approximation to $\sqrt{7}$.

```
In[4]:= Normal[%]
```

```
Out[4]=  $\frac{514088}{194307}$ 
```

It is very close to $\sqrt{7}$.

```
In[5]:= N[%, 20]
```

```
Out[5]= 2.6457513110695960516
```